METRIC

DOD-B-63545/1
19 FEBRUARY 1982

MILITARY SPECIFICATION SHEET

BOLT, RIBBED SHOULDER, WHEEL MOUNTING, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bolts described herein shall consist of this specification and the latest issue of DOD-B-63545.

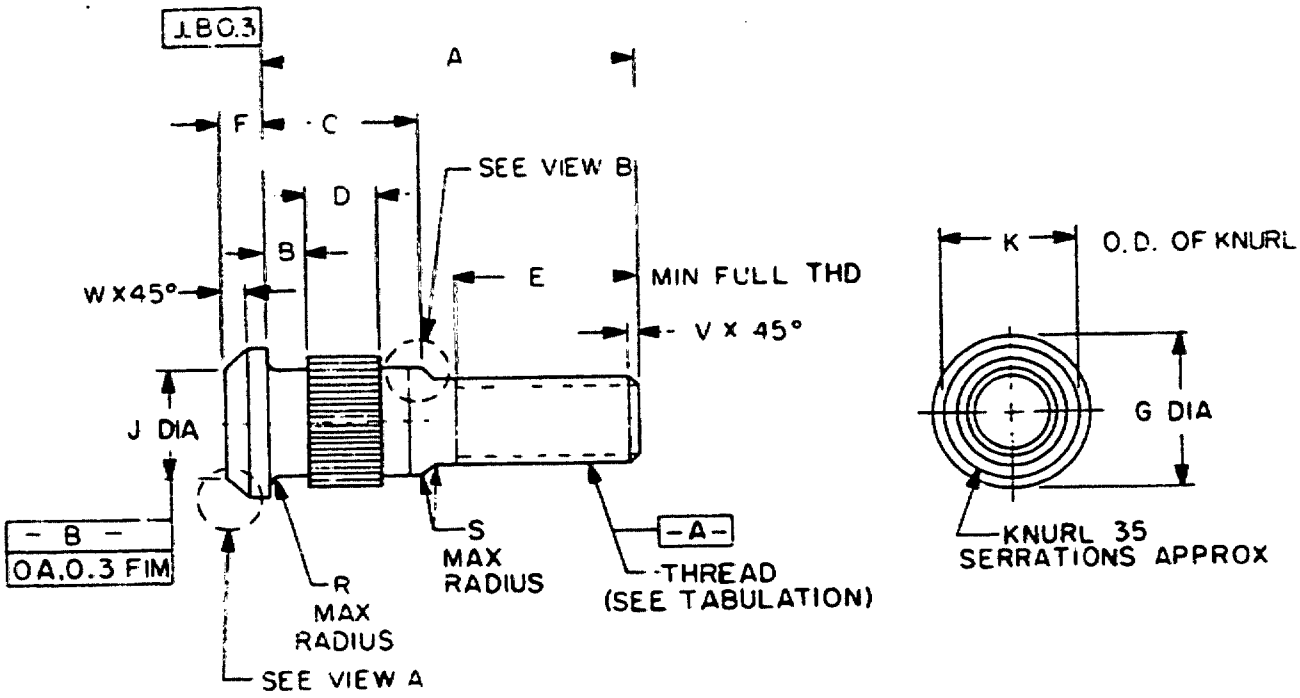


FIGURE 1. Bolt, ribbed shoulder, wheel mounting.

FSC 5306

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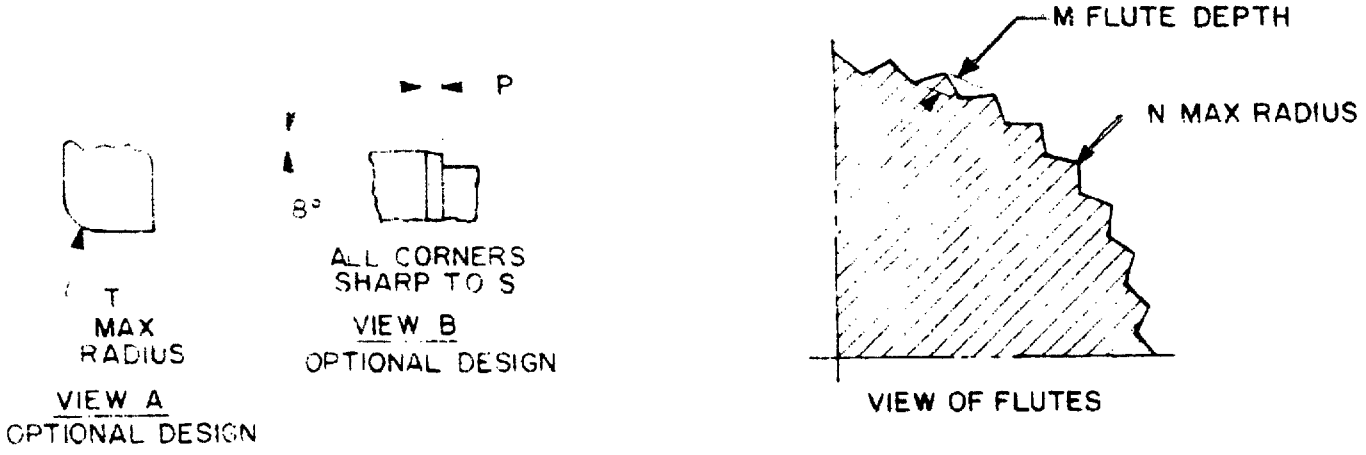


FIGURE 2. Optional views.

NOTES:
1. This standard is not intended to limit construction to features other than as shown herein by dimensions, notations, and referenced documents.
2. Tolerance on angles $\pm 1^{\circ}$

REQUIREMENTS:

1. Material. Steel alloy 4135, 4137, 4140, 5135, 5140, 8637, 8640, 8740 per FED-STD-66.
2. Hardness. Bolts shall have a hardness of 33-39HRC. Restore surface carbon if necessary to produce a surface hardness of HR15N-78-84 or equivalent. Heat treatment shall not be accomplished in presence of nitrogen compounds.
3. Material from root of serrations must not burr or increase J diameter.
4. Part number. The part number consists of the letter "D" followed by the basic number of this specification sheet, and a dash number taken from table I. Add "Z" after dash number for zinc plating. Add "C" after dash number for cadmium plating.

   Example: D63545/1-1Z = M12X1.5 bolt, zinc plated.
   D63545/1-2C = M20X1.5 bolt, cadmium plated.

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TABLE I. Dash numbers and dimensions.

Dash no.	Dia and thread	A	B	C	D	E	F	G	J	
									Max	Min
-1	M12X1.5-6g	43 ±1.0	2	8	4	30	4	16	12.44	12.33
-2	M20X1.5-6g	100 +1.5 -0.8	7.50	26 ±0.5	12.0	45	10.0	32 ±1.5	22.58	22.50
-3	M22X1.5-6g	113 +1.5 -0.8	7.50	26 ±0.5	12.0	50	10.0	32 ±1.5	24.58	24.50

TABLE I. Dash numbers and dimensions. - Continued

Dash no.	Dia and thread	K	M		N	P	R	S	T	V	W
			Max	Min							
-1	M12X1.5-6g	12.85	0.4	0.2	0.2	1.7	1.0	0.8	3	0.8	1.5
-2	M20X1.5-6g	23.60 ±0.2	1.3	0.9	0.2	2.0	1.5	1.0	5	4.0	5.0
-3	M22X1.5-6g	25.60 ±0.2	1.3	0.9	0.2	2.0	1.5	1.0	5	4.0	5.0

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing activity:
Army - AR

(Project 5306-0692)

Review activities:
Army - AT
Navy - SH, MC
DLA - IS
NSA - NS

User activities:
Army - ME
Navy - OS

Agent:
DLA - IS

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DOCUMENT IDENTIFIER (Number) AND TITLE

DOD-B-63545/1
BOLT, RIBBED SHOULDER, WHEEL MOUNTING, METRIC

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

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SUBMITTED BY (Printed or typed name and address -- Optional)

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EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

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